Table 1: AUC scores of detectors on text generated by different methods. "–" refers to the detector's AUC score on the original AI-generated text, without applying any evasion methods. Symbol '*' represents that SICO uses GPT3-D as the proxy detector for prompt construction. For each detector, the lowest AUC score is indicated in **bold**, and the second-lowest is underlined.

Dataset	Method	GPT3-D*	GPT2-D	GPTzero	OpenAI-D	DetectGPT	Log-Rank
Writing	–	0.908	0.848	0.779	0.789	0.834	0.914
	Parrot	0.666	0.645	0.632	0.744	0.502	0.577
	DIPPER	0.736	0.907	0.689	0.750	0.550	0.684
	GPT-Para	0.879	0.623	0.631	0.690	0.569	0.713
	Human Prompt	0.852	0.560	0.491	0.655	0.676	0.759
	SICO-Para	**0.239**	0.332	0.290	0.488	**0.149**	**0.147**
	SICO-Gen	0.242	**0.099**	**0.184**	**0.311**	0.441	0.318
QA	–	0.981	0.906	0.923	0.781	0.876	0.956
	Parrot	0.922	0.837	0.849	0.698	0.689	0.806
	DIPPER	0.888	0.962	0.869	0.722	0.604	0.782
	GPT-Para	0.956	0.797	0.811	0.699	0.640	0.782
	Human Prompt	0.912	0.625	0.791	0.656	0.662	0.757
	SICO-Para	**0.407**	0.576	0.572	0.541	**0.178**	**0.183**
	SICO-Gen	0.668	**0.489**	**0.494**	**0.524**	0.497	0.535
Review	–	0.925	0.952	0.939	0.960	0.808	0.982
	Parrot	0.871	0.934	0.913	0.882	0.654	0.893
	DIPPER	0.875	0.984	0.888	0.824	0.515	0.814
	GPT-Para	0.899	0.851	0.833	0.925	0.542	0.864
	Human Prompt	0.839	0.610	0.856	0.858	0.619	0.851
	SICO-Para	0.465	**0.264**	0.599	**0.540**	**0.270**	**0.300**
	SICO-Gen	**0.455**	0.619	**0.399**	0.607	0.485	0.583

For each task, we evaluate AUC score using 200 human-written text and 200 original or paraphrased AI-generated text. For each task input, we run each evasion method *only once*, instead of repeating multiple times until successful evasion, to simulate real-world scenarios where the target detector is inaccessible.

Experimental settings We set $|\mathbf{X}_{eval}| = 32$, $K = 8$, $N = 6$, and use GPT-3.5, specifically *gpt-3.5-turbo-0301*, as the LLM, where the inference parameters are kept in default. And we use GPT3-D as the proxy detector. Experiments using other LLMs and proxy detectors are presented in Section 5.2.

4.2 Evasion Performance and Analysis

Table 1 presents the performance of SICO and other baselines against six detectors in AUC score. SICO consistently outperforms other baselines by a substantial margin in all cases. Notably, in most cases, SICO reduces the AUC score to less than 0.5, equivalent to the expected performance of a random classifier. Figure 2 shows the ROC curves of evasion methods on academic writing task. One can clearly observe that SICO curves lie below others along different thresholds, often lower than the random classifier curve. More evasion results including ROC cures and detection rates are shown in Appendix G.

One interesting trend is that SICO-Para consistently outperforms SICO-Gen against statistical detectors, i.e., DetectGPT and Log-Rank. We speculate this performance difference comes from the varying influence of the prompt on the generated text between the two methods. In SICO-Para, the distribution of generated text is largely influenced by the original AI-generated text, which is in the prompt. However, in SICO-Gen, the distribution of generated text depends more on the previously generated text. Given that statistical detectors have access to the newly generated text but not the prompt, their estimation of token probability becomes

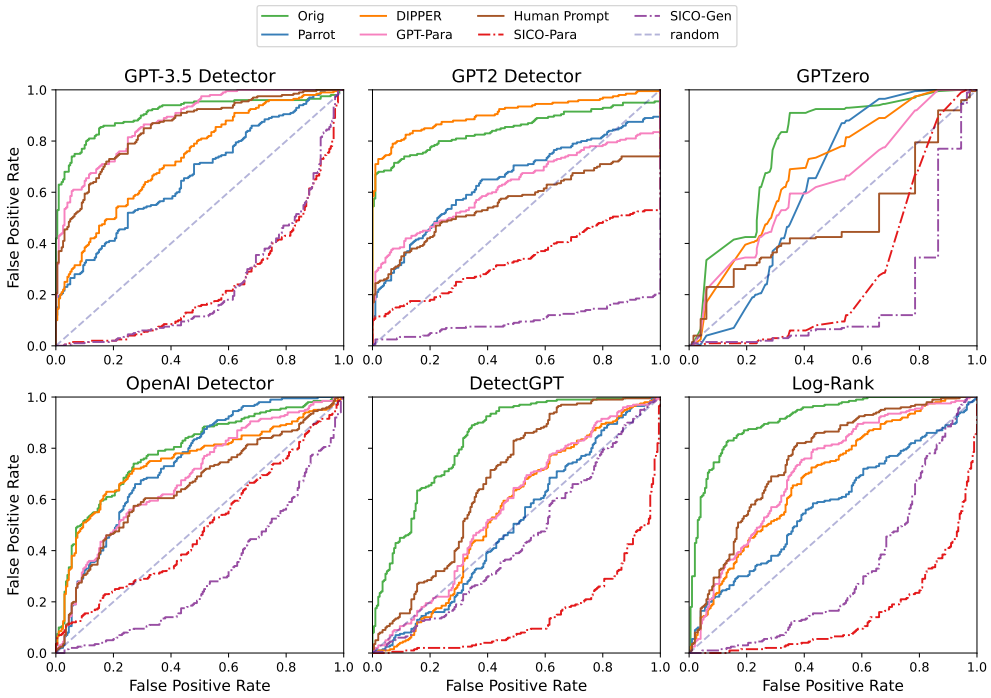

Figure 2: ROC curves for six detectors evaluating text generated by various evasion methods in an academic writing task.

less accurate for SICO-Para text, thus misleading the detection. It might also explain why GPT-Para can reduce the performance of statistical detectors.

4.3 Human Evaluation

4.3.1 Text Quality

From the users' perspective, using AI-generated text goes beyond evading detection systems; the usability of text is equally critical. For example, for academic writing task, users expect the text to be readable, properly formatted, and relevant to the given topic. Therefore, we evaluate the usability of text based on two criteria: readability and task completion rate. For each task, we randomly sample 200 examples generated by four methods (50 per method), including human-written text. Then we ask three human annotators to rate the readability of text on a scale from 1 to 5, and judge if the text accomplishes the task's goal. More details of human evaluation are shown in Appendix D.

Table 2: Human evaluation results. "Avg.D." represents the average difference between the results achieved by the evasion method and the results achieved by human-written text on the three tasks. The best value of each task is set **bold**.

Method	Readability ↑				Task Completion Rate % ↑			
	Writing	QA	Review	Avg.D. ↑	Writing	QA	Review	Avg.D. ↑
DIPPER	3.52	4.12	3.42	-0.27	70.6	100	61.6	-13.3
SICO-Para	3.68	**4.36**	3.58	-0.09	82.0	100	72.4	-5.9
SICO-Gen	3.84	4.28	**3.70**	**-0.02**	93.6	100	69.6	**-2.9**
Human-Written	**3.92**	**4.36**	3.60	-	**98.2**	100	**73.8**	-

As shown in Table 2, both SICO-Gen and SICO-Para demonstrate superior performance over DIPPER in terms of task completion and readability over three tasks. Furthermore, SICO-generated text performs competitively compared with human-written text in both metrics, with a negligible difference less than 0.1. In contrast, DIPPER exhibits inferior performance relative to human-written text, particularly with a notable 0.27 decline in readability.

4.4 Imperceptibility

Another key factor for the usability of AI-generated text is its imperceptibility. If text is easily identified as AI-generated by humans, its usability might be significantly influenced. Thus, we conducted an experiment to assess the imperceptibility of text. We sampled 200 examples (50 each from AI, DIPPER, SICO-Gen, and human) across three tasks. Three human annotators were asked to identify whether the text was AI-generated or human-written. All annotators in the experiment used ChatGPT before. The values in Table 3 show the percentage of texts recognized as AI-generated by the annotators, where a lower percentage indicates higher imperceptibility. The results demonstrate that SICO remarkably reduces the probability of being recognized by annotators for QA and review generation tasks. For the academic writing task, the similar low detection rates for AI and human-generated text (24% VS 18%) annotators' detection capabilities are less accurate for this task, explaining why SICO is less effective here. More details can be found in Appendix D.

Table 3: Imperceptibility of text from different sources.

Source	Writing	QA	Review
AI	24%	52%	54%
DIPPER	31%	46%	66%
SICO	22%	14%	37%
Human	18%	13%	22%

4.5 Cost Efficiency

In terms of the data prerequisite, SICO only needs $K + |\mathbf{X}_{eval}|$ human-written input-output examples to build prompt, which is $8 + 32 = 40$ in the experiments. The other AI-generated text can be produced by LLM leveraging these human samples. Furthermore, SICO offers the advantage of low cost for prompt construction. Based on three repeated runs, the actual USD costs of SICO-Para are 1.04 ± 0.04, 1.08 ± 0.05, and 0.75 ± 0.04 for Writing, QA, Review tasks, respectively.

5 Further Experiments

5.1 Ablation Study

We conducted an ablation study over academic writing task to to evaluate the contribution of individual components within the SICO framework. "Human-ICE" denotes the approach where human-written text is directly utilized as the in-context example for constructing the prompt. "w/o feature" and "w/o ICE" refer to the prompts without feature text and the optimized in-context examples, respectively. "w/o OPT" represents the initial prompt before optimization (see Line 3 in Algorithm 1). In our experiment, we explore SICO-Para on three types of detectors: GPT3-D, OpenAI-D and DetectGPT.

Results in Table 4 shows that directly using human-written text is ineffective, even making the detection more accurate. We speculate that the human-written examples are too heterogeneous and characterized in multiple ways, so LLM cannot effectively learn their attributes. Besides, the importance of feature text is comparatively less than that of optimized in-context examples. Furthermore, the result reveals the significant role of the optimization step in SICO. Using in-context examples that are not optimized is essentially equivalent to not using any in-context examples.

5.2 SICO with Different Proxy Detectors and LLMs

As described in Section 3, SICO requires a proxy detector and a LLM to construct a prompt. In this experiment, we explore the performance of SICO-Para on writing task, using three types of proxy detectors: (1) training-based model GPT-3.5 detector, (2) API detector GPTzero, and (3) statistical method DetectGPT.